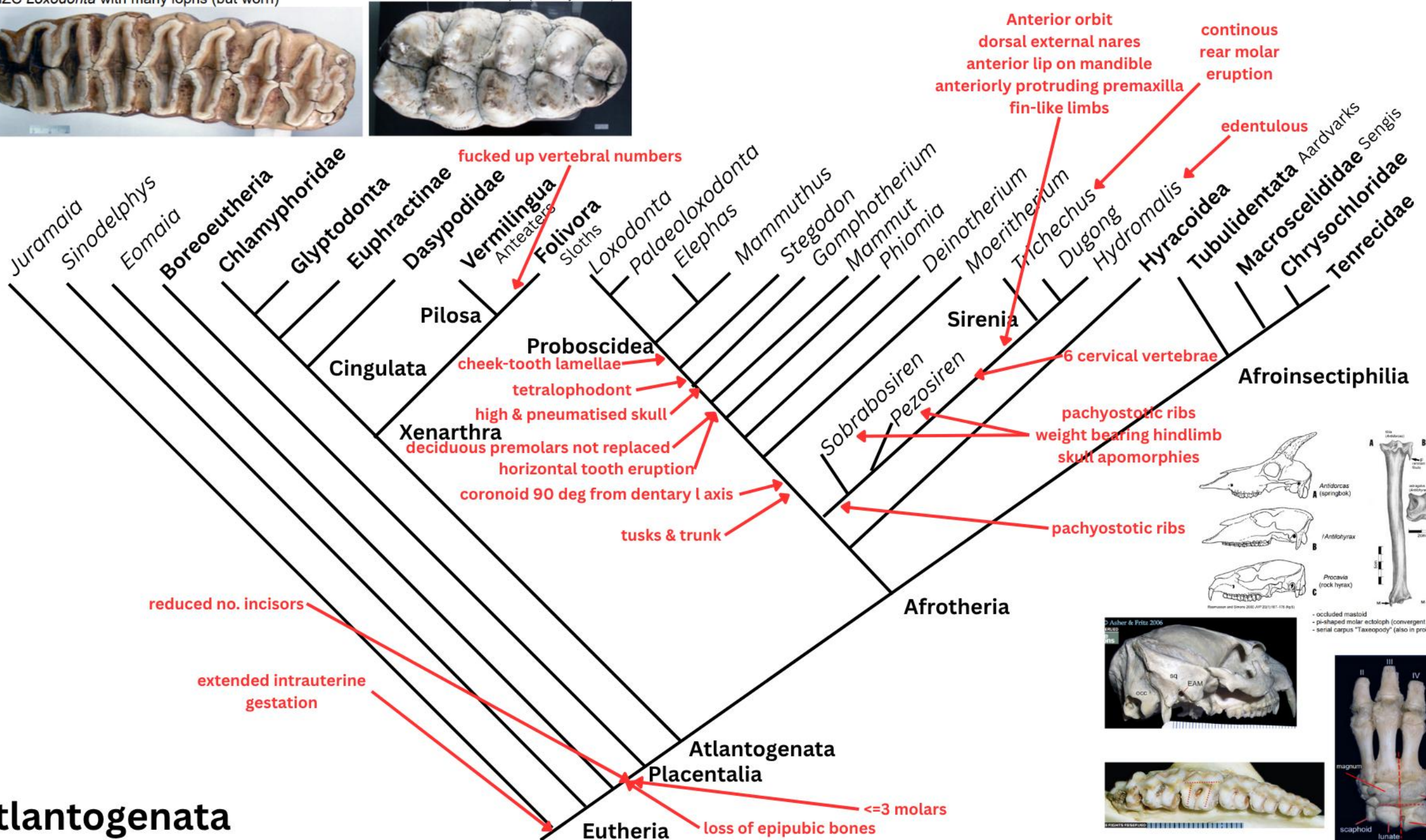
UMZC *Loxodonta* with many lophs (but worn)

UMZC *Mammut* with fewer lophs (relatively unworn)



Atlantogenata

Proboscidea

Key Adaptations

- Tusks & trunk
- Horizontal tooth eruption
- Deciduous premolars (not-replaced)
- High, pneumatised skull
- Coronoid 90 deg to long axis of dentary
- Upper diastema
- Anterior orbit
- EAM (external auditory meatus) bound by squamosal
- **Tetralophodont** dentition

Key take-aways from their evolution:

- General trend to larger teeth with more lophs
- Highly mosaic evolution of traits
- Many clades specialised and diversified and went extinct without contributing to modern proboscidean taxa
- Extremely successful group on all continents
- Lots of convergence
- General loss of tusks in modern species (smaller/no tusks)
- Early members of the group had 4 tusks (phiomia + gomphotheres)

Elephas (Asian Elephant)	Loxodonta (African Elephant)
Smaller	Larger
Smaller ears	Larger “fan” ears
Dorsally convex thoracic region	Dorsally concave thoracic region
1 finger on trunk	2 fingers on trunk
Curled enamel plates	Flatter enamel plates

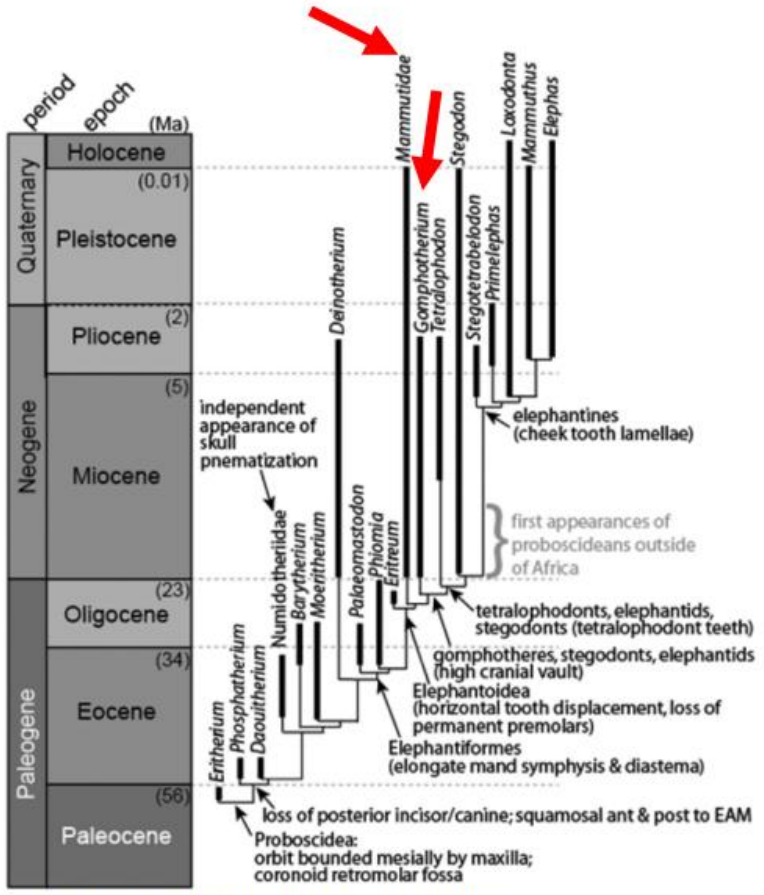
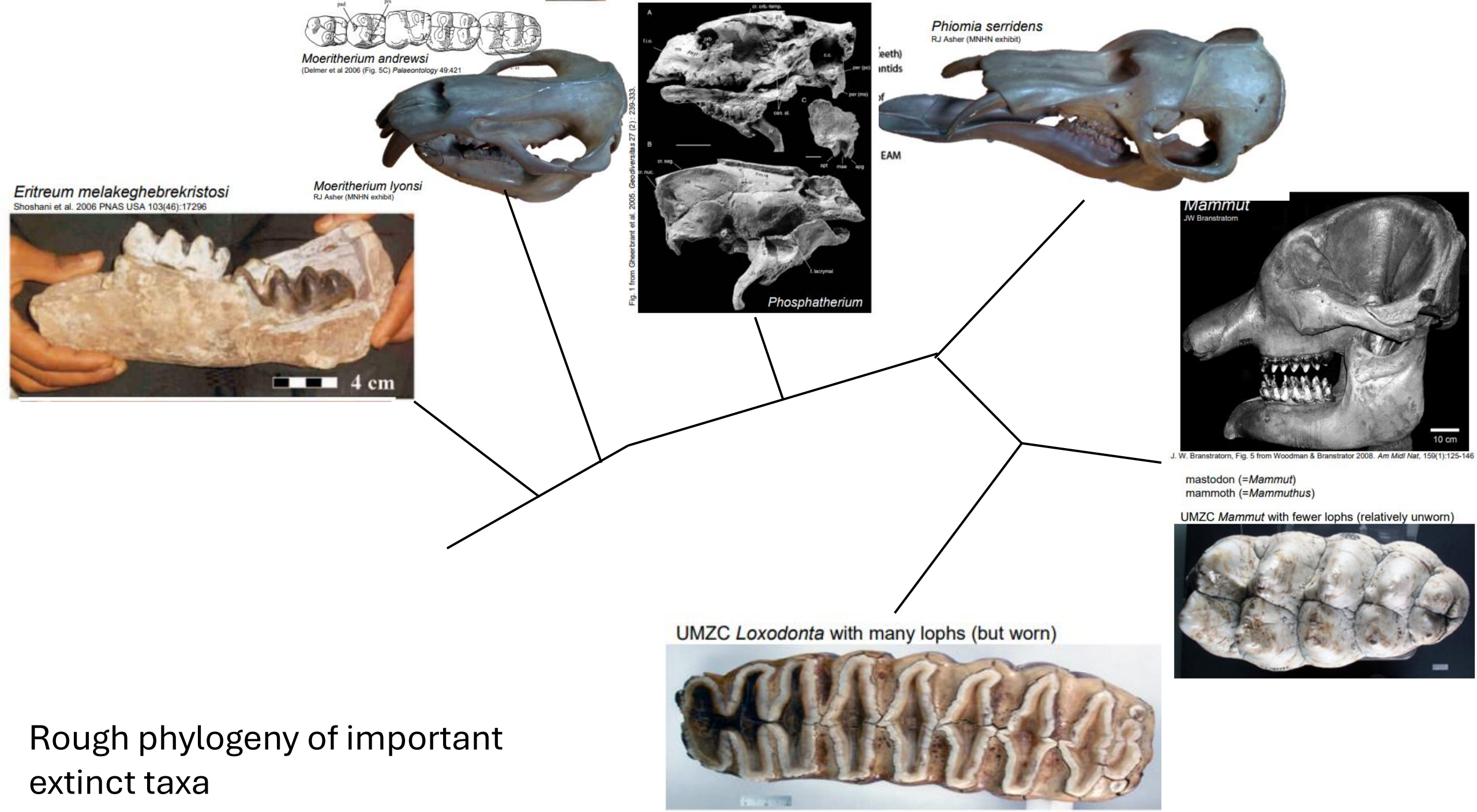


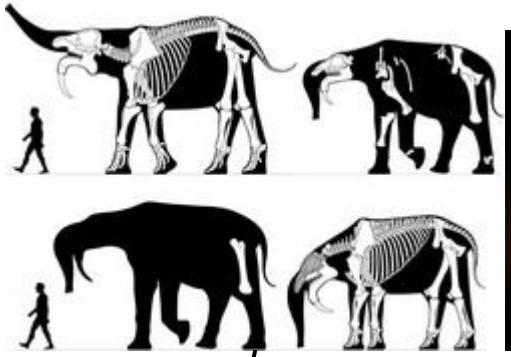
Fig. 6.2 from Asher RJ 2012 *Evolution & Belief* Cambridge Univ Press



Rough phylogeny of important extinct taxa

Other Cool Taxa

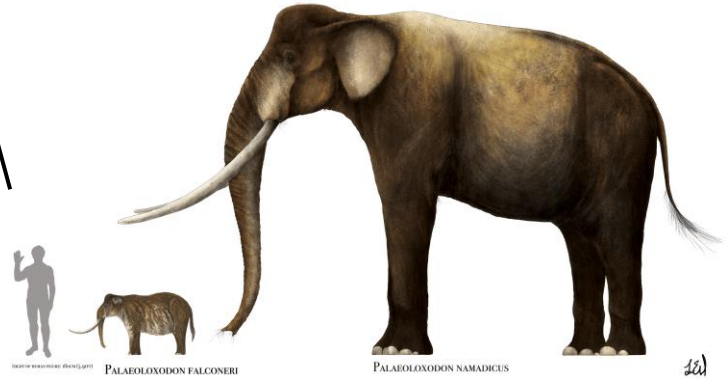
Deinotherium



Mastodon (Mammut)

Gomphotherium

Loxodonta



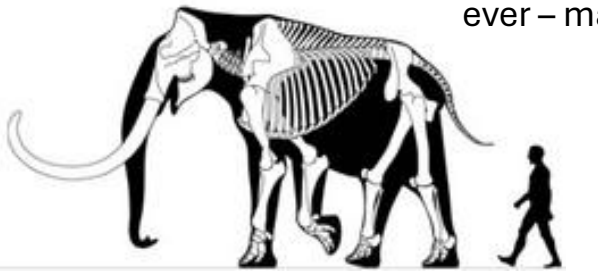
Paleoloxodon
(Largest terrestrial mammal ever – maybe)

Elephas

Steppe Mammoth

Woolly Mammoth

Columbian Mammoth



Afroinsectiphilia

Ungulates

- Large body size
- Herbivorous
- 4 cusped upper cheek teeth
- Digitigrade
- Flattened terminal digits

Faunivores

- small body size
- faunivorous
- 3 cusped upper cheek teeth
- plantigrade
- narrow terminal digits

Zalambodonty

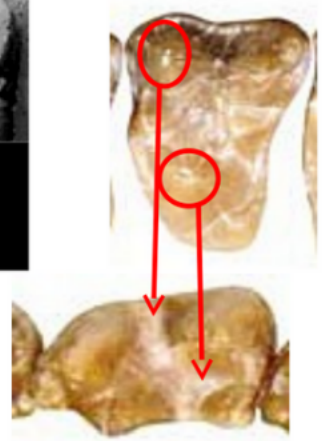
Upper molars comprised of "V" with main cusp paracone*
Lower molars show reduced/absent talonid basin

Asher RJ, Sánchez-Villagra MR. 2005. *J Mamm Evol* 12 (1/2): 265-282.

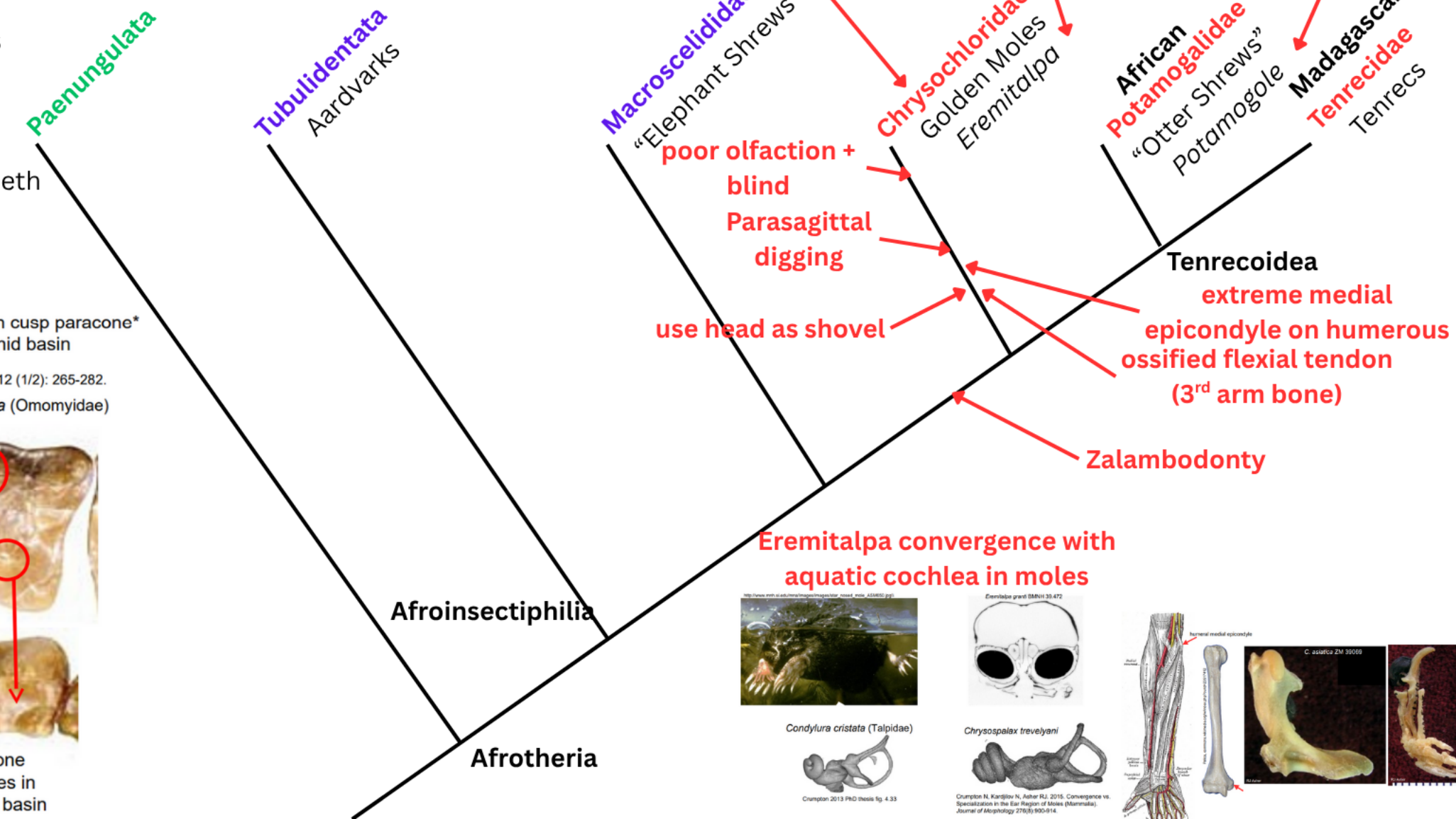
Limnogale (Tenrecidae)

paracone occludes in ectoflexid

Teilhardina (Omomyidae)



protocone occludes in talonid basin



Faunivore-Ungulate Intermediates within Afroinsectiphilia

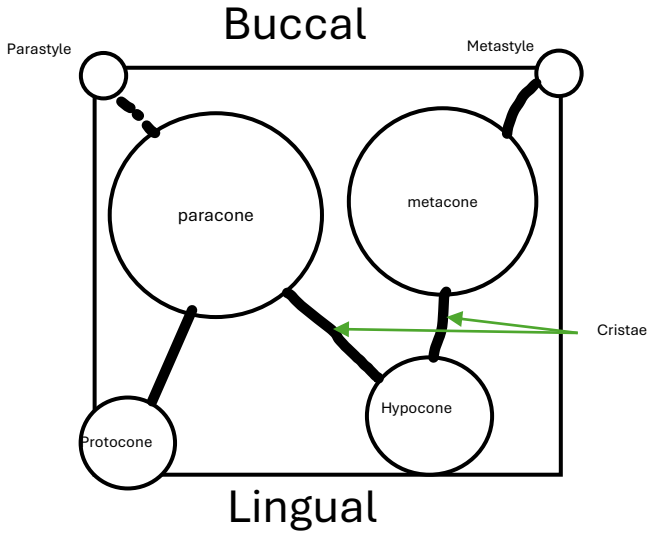
Tubulidenta (aardvarks):

- Faunivorous
- Fossorial (sort of)
- Large body size
- Unguligrade locomotion (tip-toed-hoofed running)

Macroscelididae (elephant shrews/sengis):

- Small
- Faunivorous
- Narrow terminal digits
- Quadritubercular cheek teeth (4 main cusps – paracone, protocone, metacone, AND hypocone)
- Unguligrade locomotion

Quadritubercular tooth with Zalambodonty



True Faunivores within Afroinsectiphilia

Tenrecidae (tenrecs):

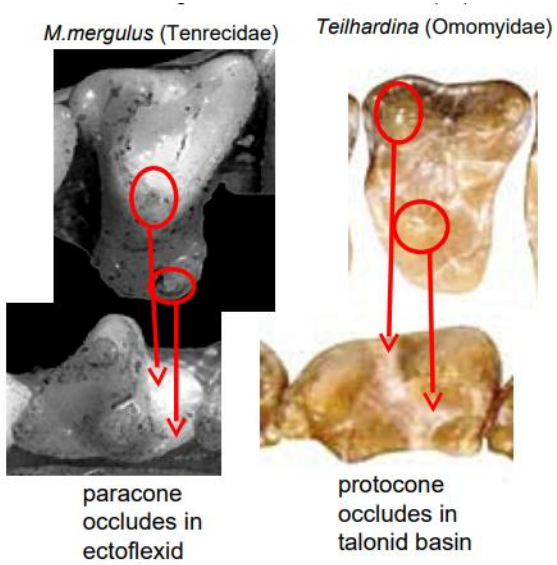
- 10 genera, 30sp, mainly Madagascan
- Look like hedgehogs
- Potamogale (African genera) is aquatic and swims with lateral undulations!

Chrysochloridae (golden moles):

- Parasagittal digging
- Look like moles

Both have Zalambodonty:

- Zalambodonty - lambda shape due to ridges between cusps
- Paracone becomes main cusp, ectoflexid enlarged, talonid basin effectively redundant/reduced.



EVIDENCE for Afrotheria

- Genetic Evidence (whole genome, developmental, mitochondrial)
- Anatomical evidence is sparse as the group has diversified so much that many have lost their defining characters.
- Testicondy
- delayed or non-existent descending of the testicles
- Present in most afrotherians including some tenrecs and chrysochlorids
- Increased number of thoracolumbar vertebrae
- Variations in vertebral count – shared with xenarthrans.

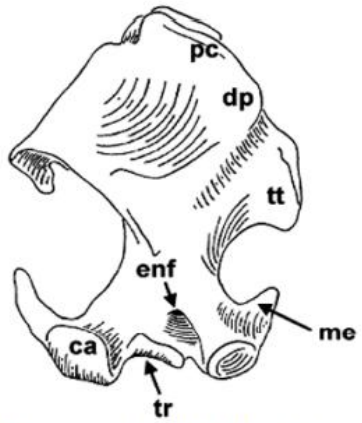
Reasons why testicondy is a bad trait to use:

- Many other mammalian groups have the same condition
- Descended but ascrotal:
- Aardvarks, tenrecines, xenarthrans, lipotyphlans, microbats, non-equid perissodactyls, cetaceans, pinnipeds
- Descended, scrotal:
- Most non-aquatic mammals in euarchontoglires and laurasiatherians.

Chrysochlorids vs Talipids

Chrysochloridae	Talpidae
Medially directed proximal ulna	Inverted orientation of humerus
Hypertrophied medial humeral epicondyle	Elongated + narrow scapula & keeled sternum + reduced clavicle
Ossifeid flexor tendon (3rd forearm bone)	Enlargement of teres major muscle for rotating humerus
Use head like a shovel for digging	Enlarged sesamoid/"6th digit"
Parasagittal digging	Lateral Scratch digging

Talpa europaea



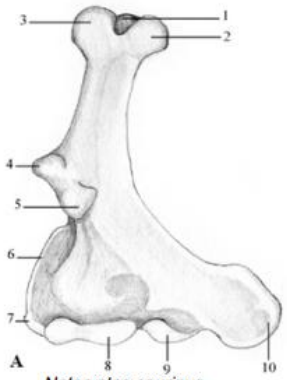
Talpa europaea (Geisler 2004: fig. 13.2)

"lateral scratch digger"

golden mole

Chrysochloris sp. (SAM PQL 67132)

marsupial mole

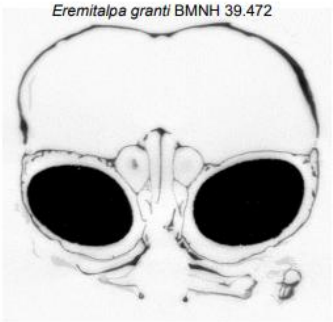


Notoryctes caurinus
(Warburton 2003: fig. 4.7)

"parasagittal head & forelimb diggers"

http://www.mnh.si.edu/mna/images/images/star_nosed_mole_ASM650.jpg\\

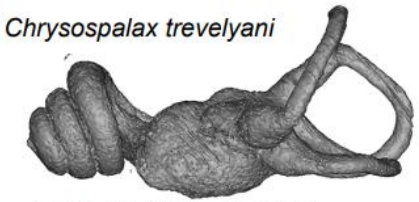
Eremitalpa granti BMNH 39.472



Condylura cristata (Talpidae)

Crumpton 2013 PhD thesis fig. 4.33

Chrysospalax trevelyani



Crumpton N, Kardjilov N, Asher RJ. 2015. Convergence vs. Specialization in the Ear Region of Moles (Mammalia). *Journal of Morphology* 276(8):900-914.

Eremitalpa (dune shark)

Main points:

- Extremely picky eater, identifying only termites through vibrations in sound alone
- Similar to an aquatic mammal in that respect
- Giant malleus
- It seems a elongate coiled cochlea improves low frequency sound detection in certain mediums and is therefore adopted by aquatic species.
- Chrysochlorids are quite limited to adapt in this way, hence why there is likely no aquatic golden moles.
- However, eremitalpa comes quite close.

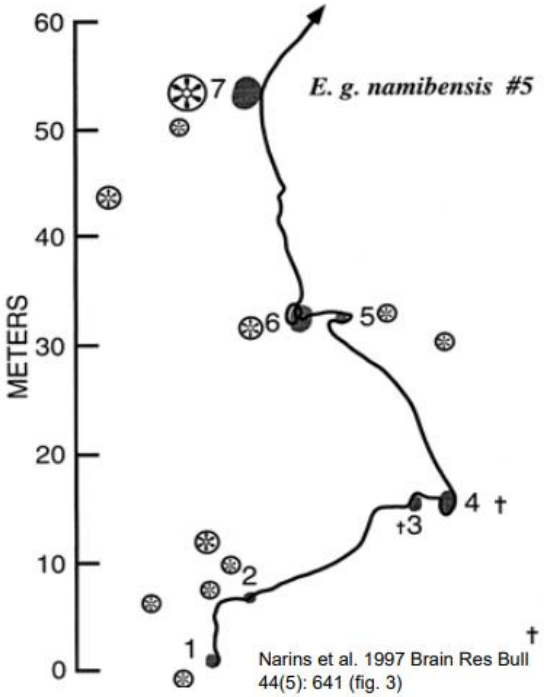
TABLE I

Dietary composition of ***E. g. namibensis*** *in comparison to prey availability at Gobabeb*

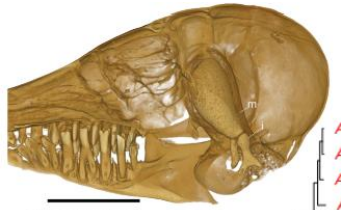
Item	Percentage composition by number: Diet ($n=16$)[a]	Field ($n=397$)[b]
Isoptera	97·5	0·2
Insect larvae	1·7	9·4
Coleoptera	0·3	27·7
Formicidae	0·2	0·0
Aranidae	0·1	5·5
Thysanura	0·1	50·1
Hemiptera	0·1	0·2
Scincidae	0·1	0·0
Hymenoptera	0·0	1·2
Mantidae	0·0	0·5
Solpugidae	0·0	0·2
Total items	2133	415

[a] Number of stomachs examined
[b] Number of excavation samples

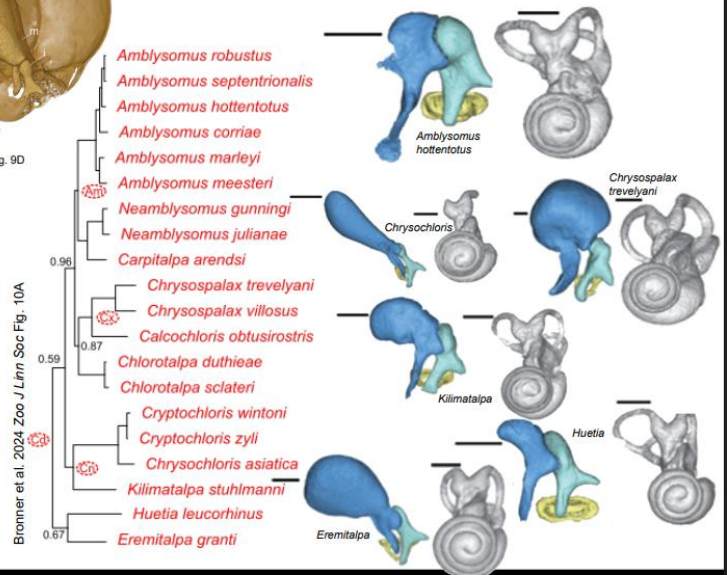
Fielden et al. 1990 J Zool London 220:367 (table 1)
Isoptera = termites
Thysanura = silverfish
Coleoptera = beetles



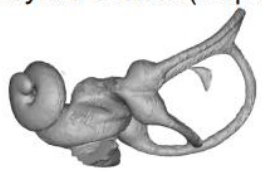
Narins et al. 1997 Brain Res Bull 44(5): 641 (fig. 3)



Cryptochloris wintoni
Bronner et al. 2024 Zoo J Linn Soc Fig. 9D

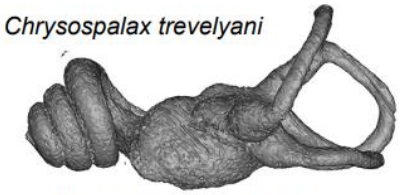


Condylura cristata (Talpidae)



Crumpton 2013 PhD thesis fig. 4.33

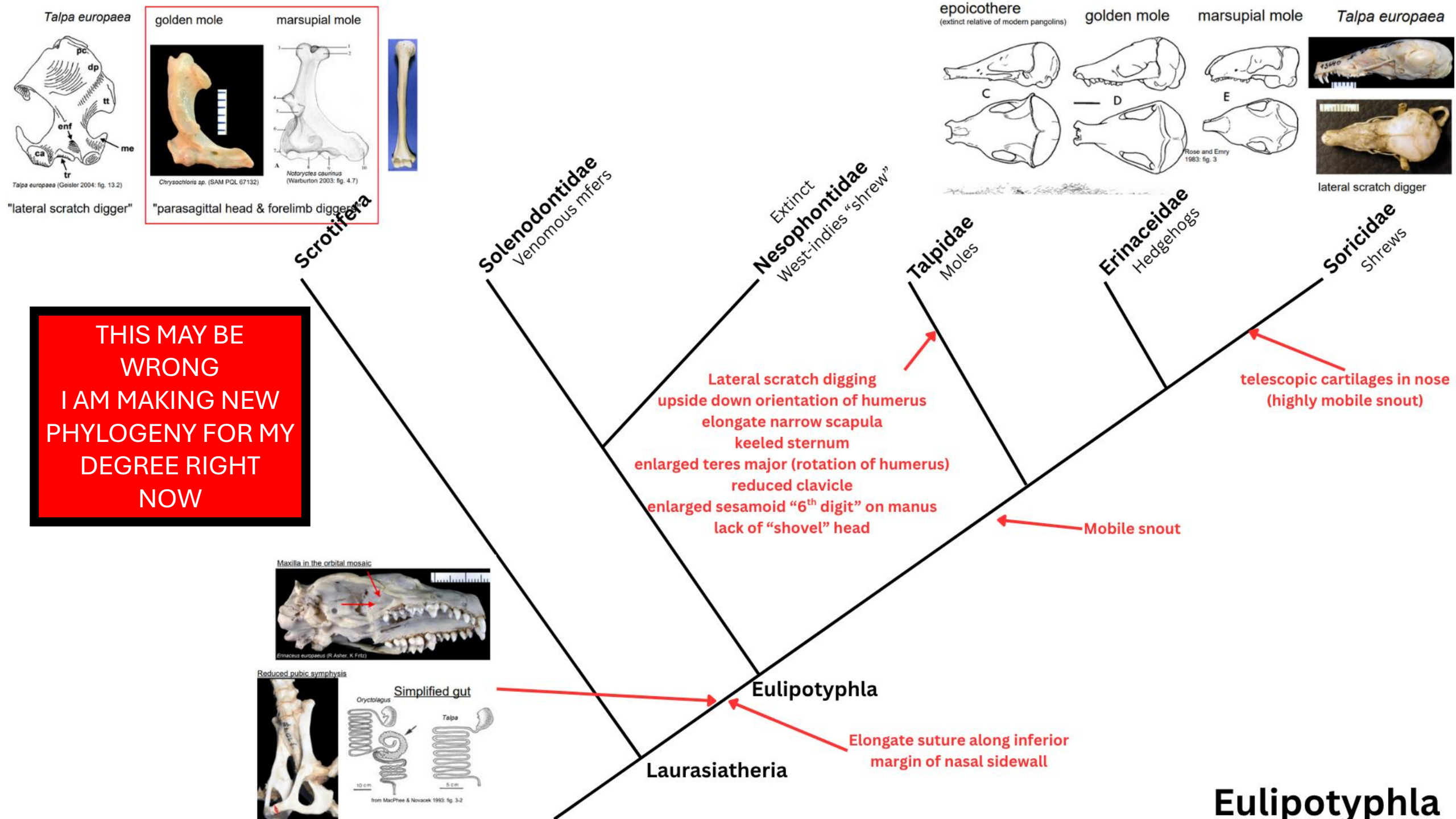
Chrysospalax trevelyani



Crumpton N, Kardjilov N, Asher RJ. 2015. Convergence vs. Specialization in the Ear Region of Moles (Mammalia). *Journal of Morphology* 276(8):900-914.

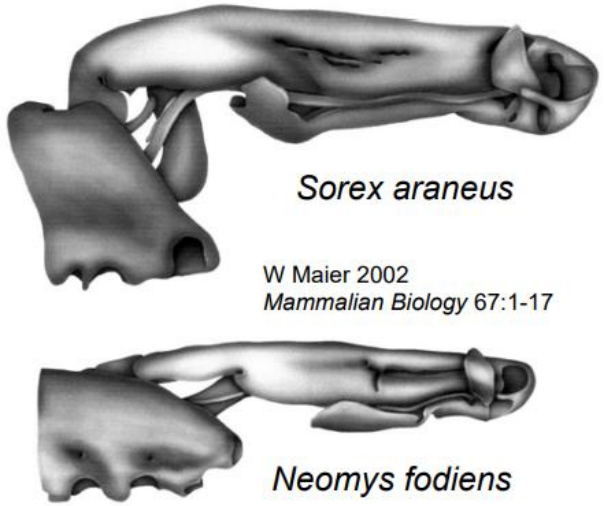
A

Ekdale 2013: fig. 56 and table 2
Cavia porcelus
1457 degree coiling!

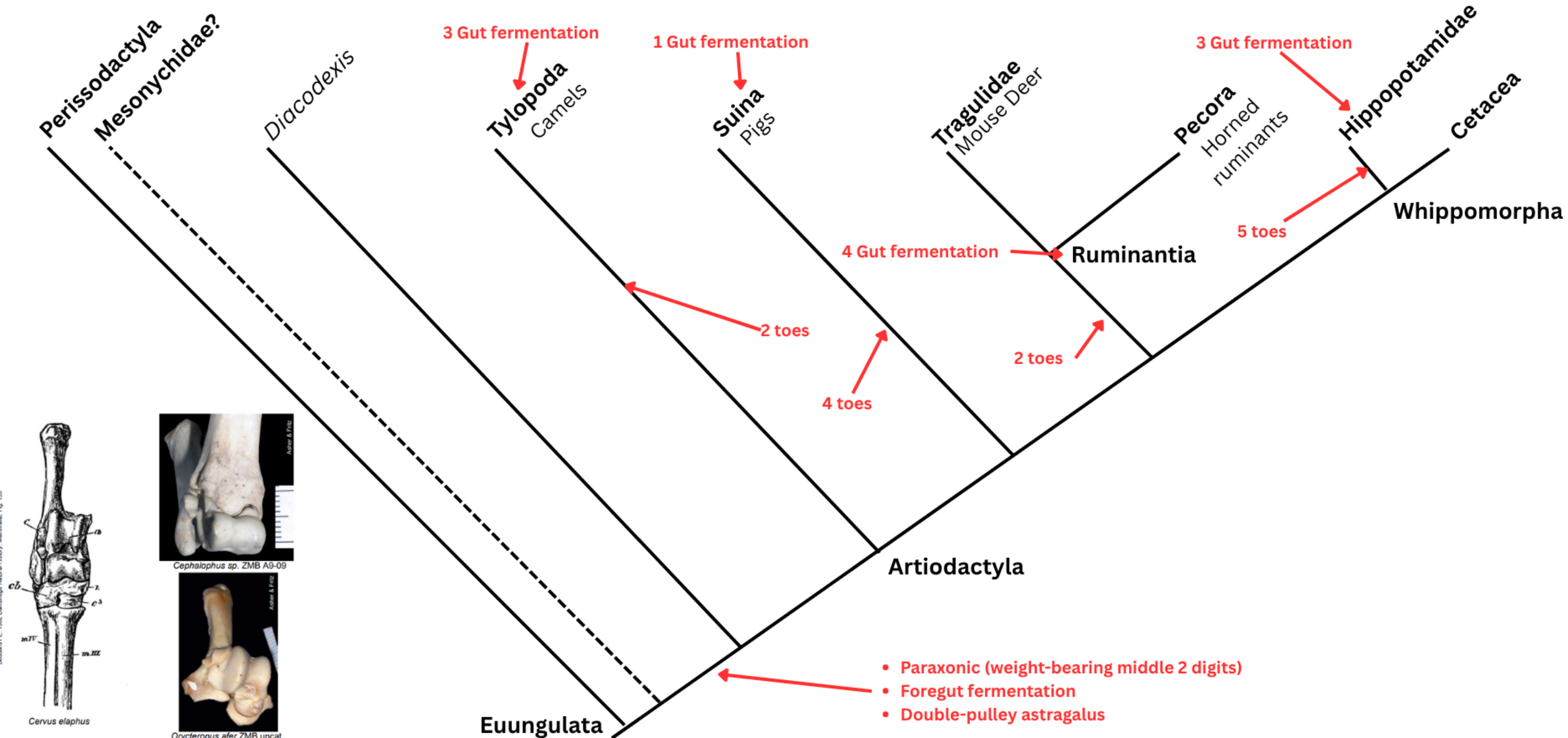

Talpa europaea
pc
dp
tt
enf
me
ca
tr
Talpa europaea (Geisler 2004: fig. 13.2)
"lateral scratch digger"
golden mole
marsupial mole
Chrysochloris sp. (SAM PQL 67132)
Notoryctes caurinus (Warburton 2003: fig. 4.7)
"parasagittal head & forelimb digger"
epoicothere
(extinct relative of modern pangolins)
golden mole
marsupial mole
Talpa europaea
C
D
E
Rose and Emry 1983: fig. 3
lateral scratch digger
Scrotifera
Solenodontidae
Venomous mfers
Extinct
Nesophontidae
West-indies "shrew"
Talpidae
Moles
Erinaceidae
Hedgehogs
Soricidae
Shrews
THIS MAY BE WRONG
I AM MAKING NEW PHYLOGENY FOR MY DEGREE RIGHT NOW
Lateral scratch digging
upside down orientation of humerus
elongate narrow scapula
keeled sternum
enlarged teres major (rotation of humerus)
reduced clavicle
enlarged sesamoid "6th digit" on manus
lack of "shovel" head
telescopic cartilages in nose
(highly mobile snout)
Mobile snout
Maxilla in the orbital mosaic
Erinaceus europaeus (R Asher, K Fritz)
Reduced pubic symphysis
Simplified gut
Oryctolagus
Talpa
10 cm
5 cm
from MacPhee & Novacek 1993: fig. 3-2
Eulipotyphla
Elongate suture along inferior margin of nasal sidewall
Laurasiatheria
Eulipotyphla

Shared derived features of lipotyphla

- Maxilla in orbital mosaic
- Reduced pubic symphysis
- Simplified gut – why?
- Cartilaginous, almost prehensile, ”bendy” rostrum.
- Some lose the zygomatic arch – not exclusive to this group though

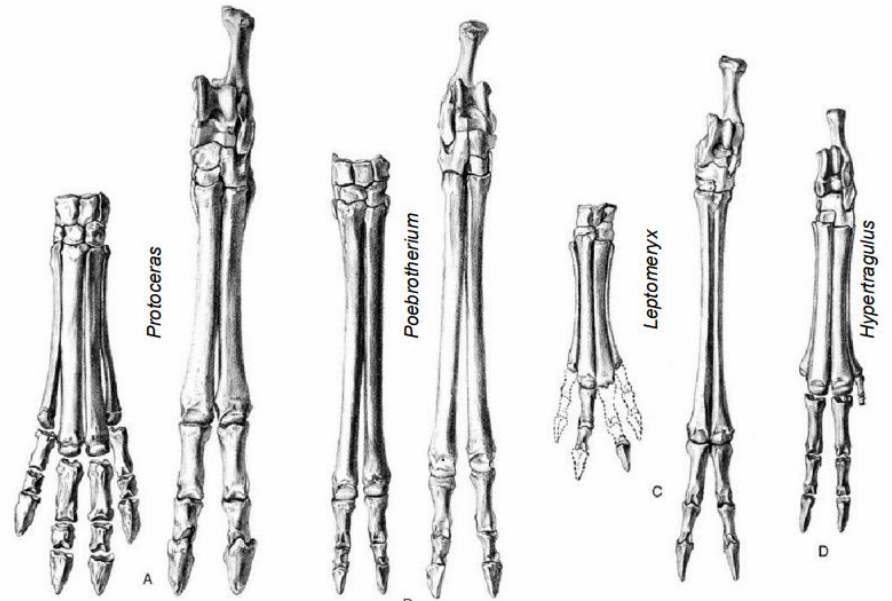


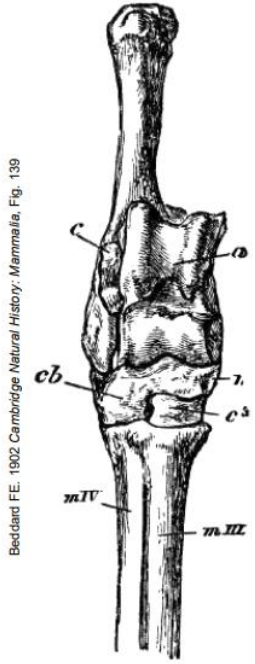
Artiodactyla



Artiodactyl adaptations

- BIG FOREGUT (fermentation)
- Paraxony (weight bearing on two middle toes)
- Double-pulley astragalus – highly efficient for cursioral locomotion due to restricting lateral movement of ankle and promoting more efficient parasagittal movement

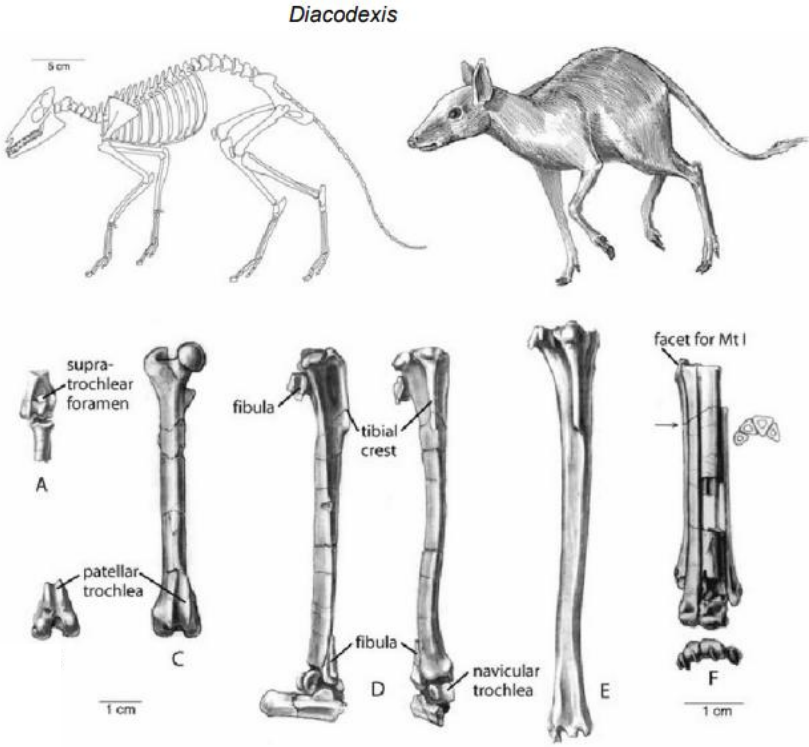




Cervus elaphus

Cephalophus sp. ZMB A9-09

Orycteropus afer ZMB uncat.



Adaptations of Cetacea

Postcranial

- Shortened neck
- No sacrum/vestigial hindlimb/pelvis
- Immobile elbow
- Hyperphalangy (>3 phalanges)

Cranial

- Telescopic skull (asymmetric in odontocetes)
- Dorsally situated external nares
- Tympanic involucrum (dense layer of bone in inner wall of tympanic bulla)
- No ear canal/plugged
- Petrosal is pachyostotic – loosely attached in cetaceans
- Odontocetes receive sound through jaw not ear
- Odontocetes have a melon for sound generation in echolocation

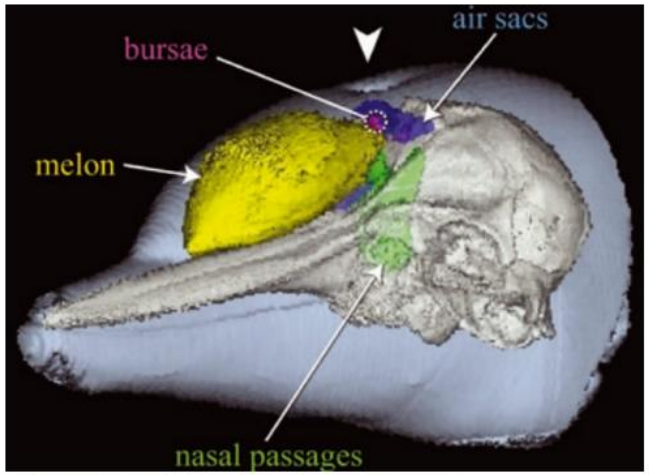


fig. 1, McKenna et al. 2012. Marine Mammal Science, 28(4): 690–713.
DOI: 10.1111/j.1748-7692.2011.00526.x

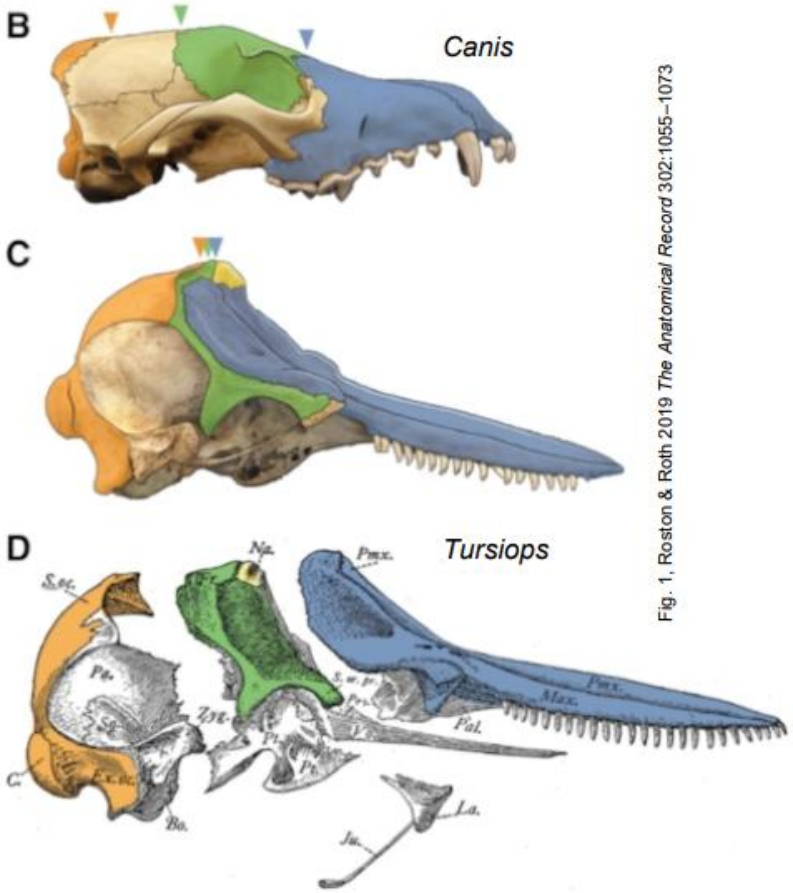


Fig. 1, Roston & Roth 2019 *The Anatomical Record* 302:1055–1073

blue = maxilla & premaxilla
green = frontal
orange = occipital
yellow = nasal

Odontoceti
(toothed whales)

- Echolocation (melon)
- Teeth (typically conical)
- Fused Mandible
- Bowed mandible
- HEAVY convergence with mosasaurs, ichthyopterygians, & thalattosuchians.
- Macropredation/piscivory/faunivory

Livyatan (macro-raptorial sperm whales)

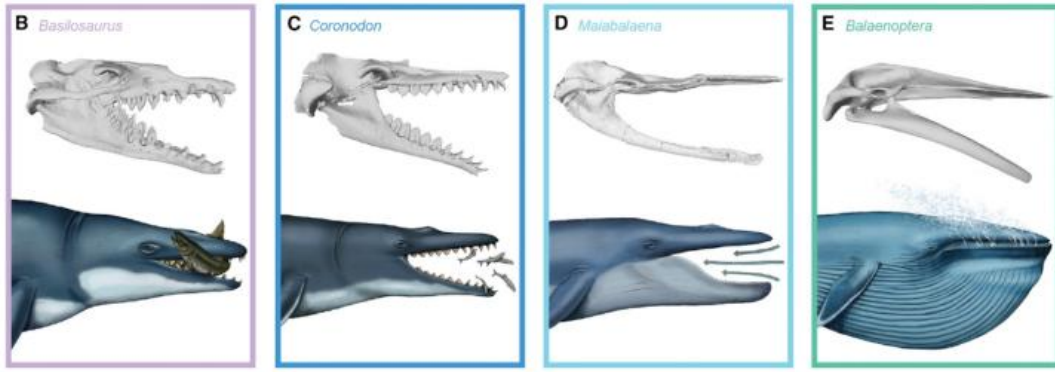
Mysticeti
(baleen whales)

- Mosaic evolution of traits
- Unfused mandibular symphysis
- Thin lateral maxilla
- Bowed mandibles
- Foramina for baleen
- BALEEN (no mineralised teeth)
- Filter feeders

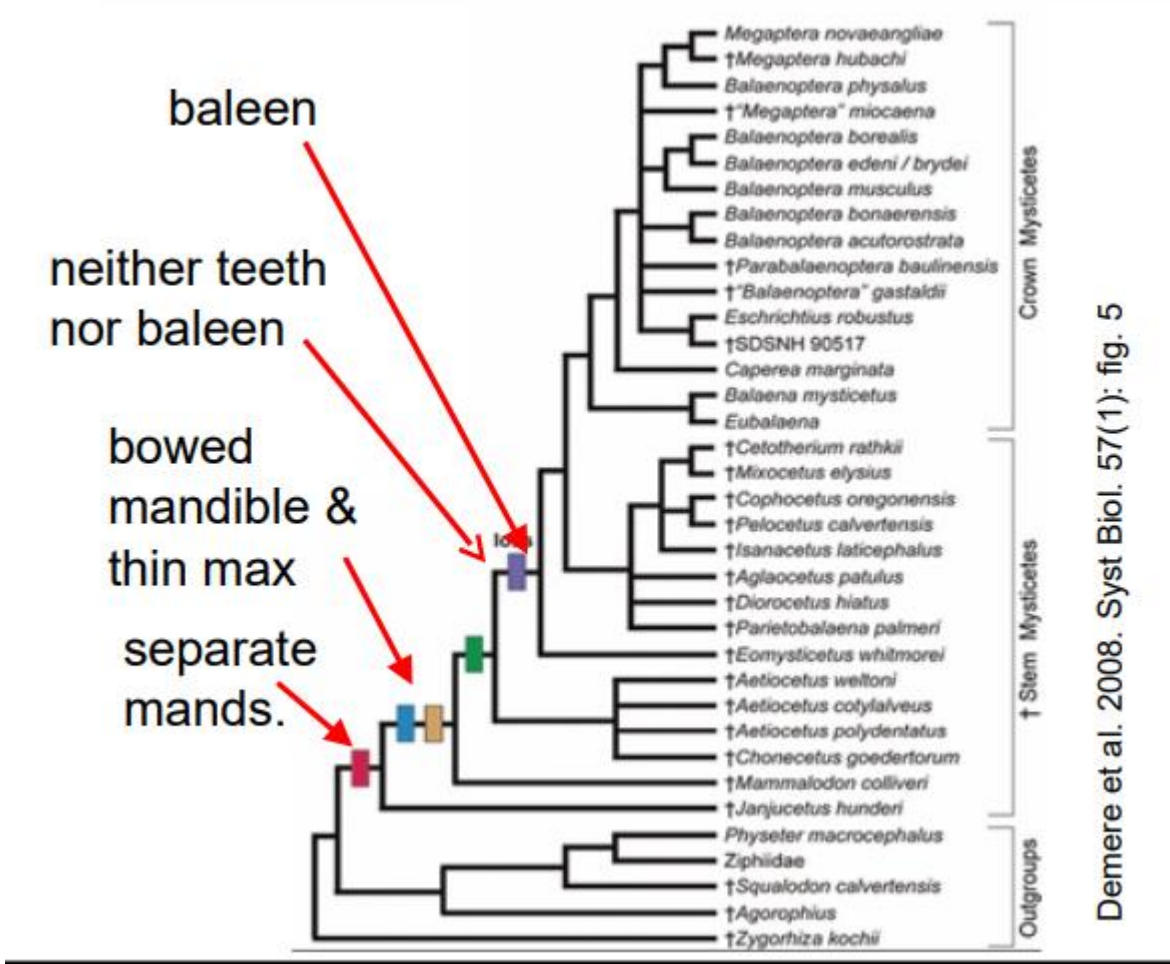
Balaenoptera acutorostrata (Modern)

Aetiocetus weltoni (Oligocene ~24-28 Ma)

Mosaic Evolution in Mysticetes



- Interpretation of palatal foramen as evidence of baleen (Demere et al. 2008).
- Peredo et al. 2018 argued tooth loss occurred before baleen – how did they feed.
- Likely fed via suction feeding – some "basilosaurids" may have also fed in this way – Perucetus.
- Tooth loss likely occurred via frameshifts in key developmental genes influencing tooth development (ENAM, AMBN, DMP1) – these are referred to as pseudogenes as they are non-functional.
- Mysticetes show frameshift in enamel generating genes but not dentine and tooth growth genes (DMP1).



Demere et al. 2008. Syst Biol. 57(1): fig. 5

Cetacea

"Archaeoceti"

Hippopotamidae
Indohyus
Pakicetus
Ambulacetus
Remingtonocetus
Maiacetus
Rodhocetus
Dorudon
Basilosaurus
Mammalodon
Aetiocetus
Eomysticetuus
Cetotherium
Balaenidae
Neobalaenidae
Balaenopteridae
Eschrichtiidae
Kogiidae
Physeteridae
Platanistidae
Ziphiidae
Inioidea
S. Am. River
Monodontidae
Phocoenidae
Delphinidae

Crown

Baleen?

Tooth Loss

Bowed Mand.
Thin Maxilla

Mysticeti

Delphinoidea

Odontoceti

Sep. Mand.

symmetric
telescopic skull

melon +
recieve sound
through jaw

asymmetric
telescopic skull

loosely
attached
petrosal

Cetacea (crown)

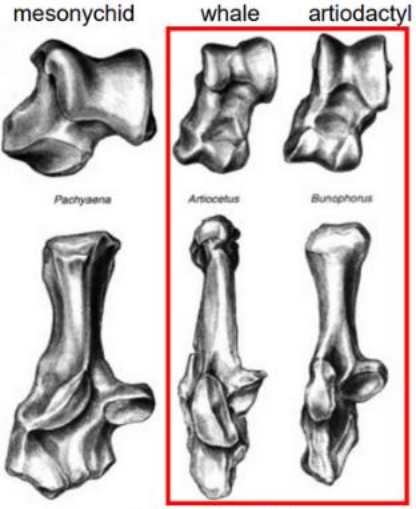
loss of hindlimbs

partial telescopic skull
dorsal ext nares
plugged ear canal
loss of sacrum

tympanic involcrum

pachostotic petrosal

Cetacea (total)



Rose 2006: fig. 14-11

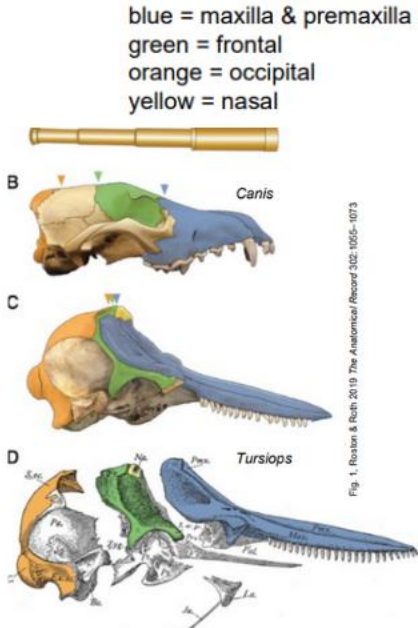
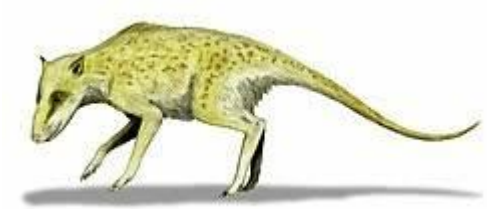


Fig. 1, Roston & Roth 2019 The Anatomical Record 302:1055–1073

Archaeocetes

Pakicetus



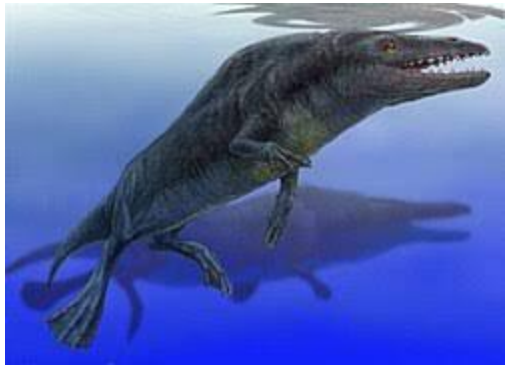
Indohyus

Largely terrestrial semi-aquatic generalists

Ambulocetus

Remingtonocetus

Rodhocetus



Semi-aquatic specialist predators

Which 4 of these do you think are the most relevant/important for understanding cetacean origins?

Maiacetus

Fully aquatic

Basilosaurus

Perucetus

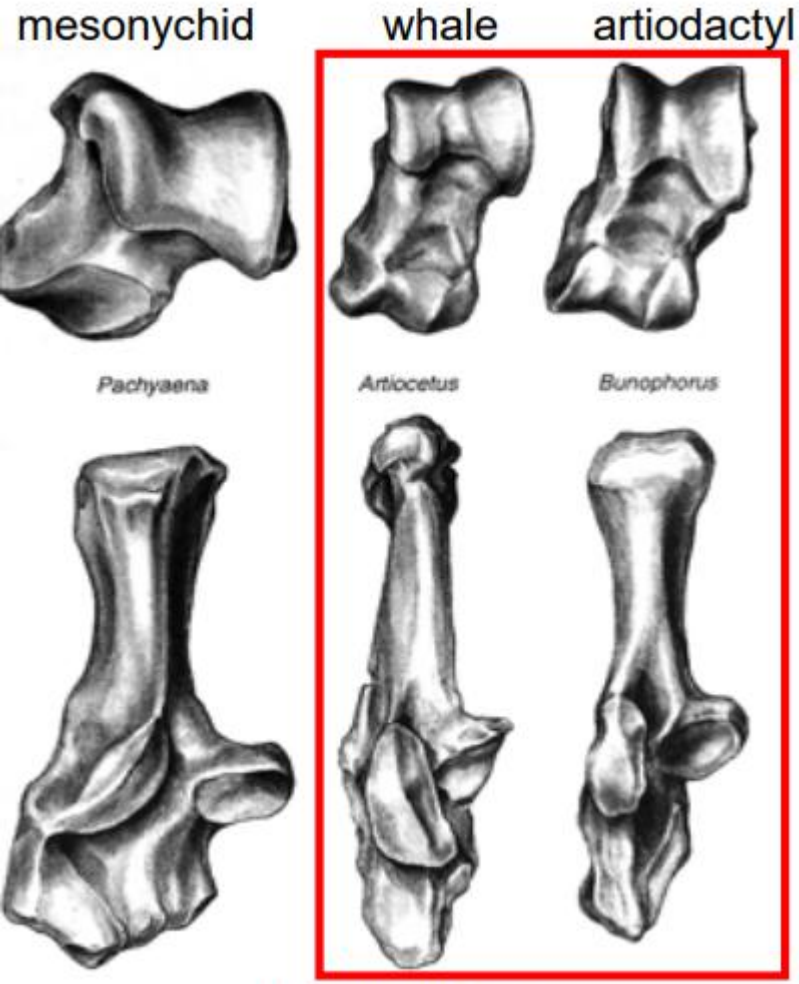
Dorudon

Cetaceans vs Mesonychids

Andrewsarchus (Mesonychid)

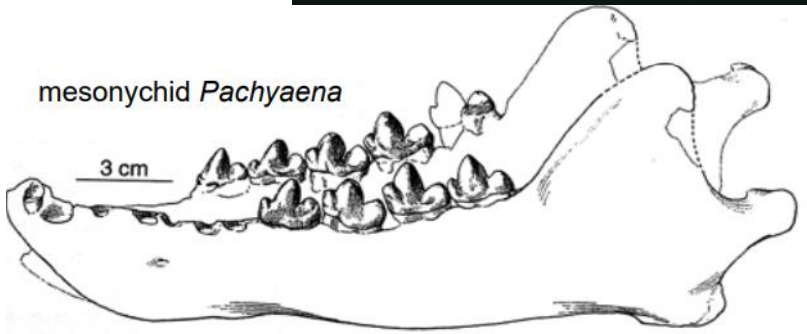
What do you immediately notice about mesonychid & archaeocete skulls?

Pakicetus (“Archaeocete”)



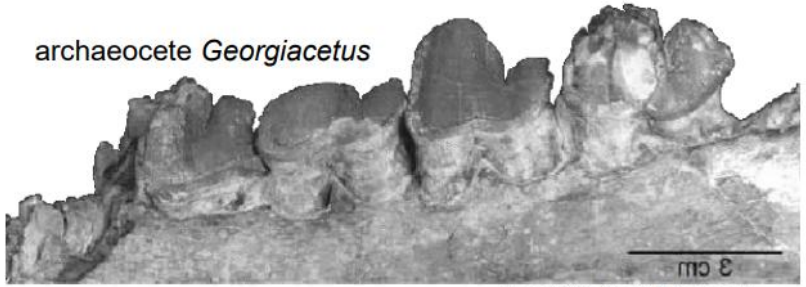
Rose 2006: fig. 14-11

Which anatomical features do you think take precedence? EXCLUDING GENOMIC DATA.

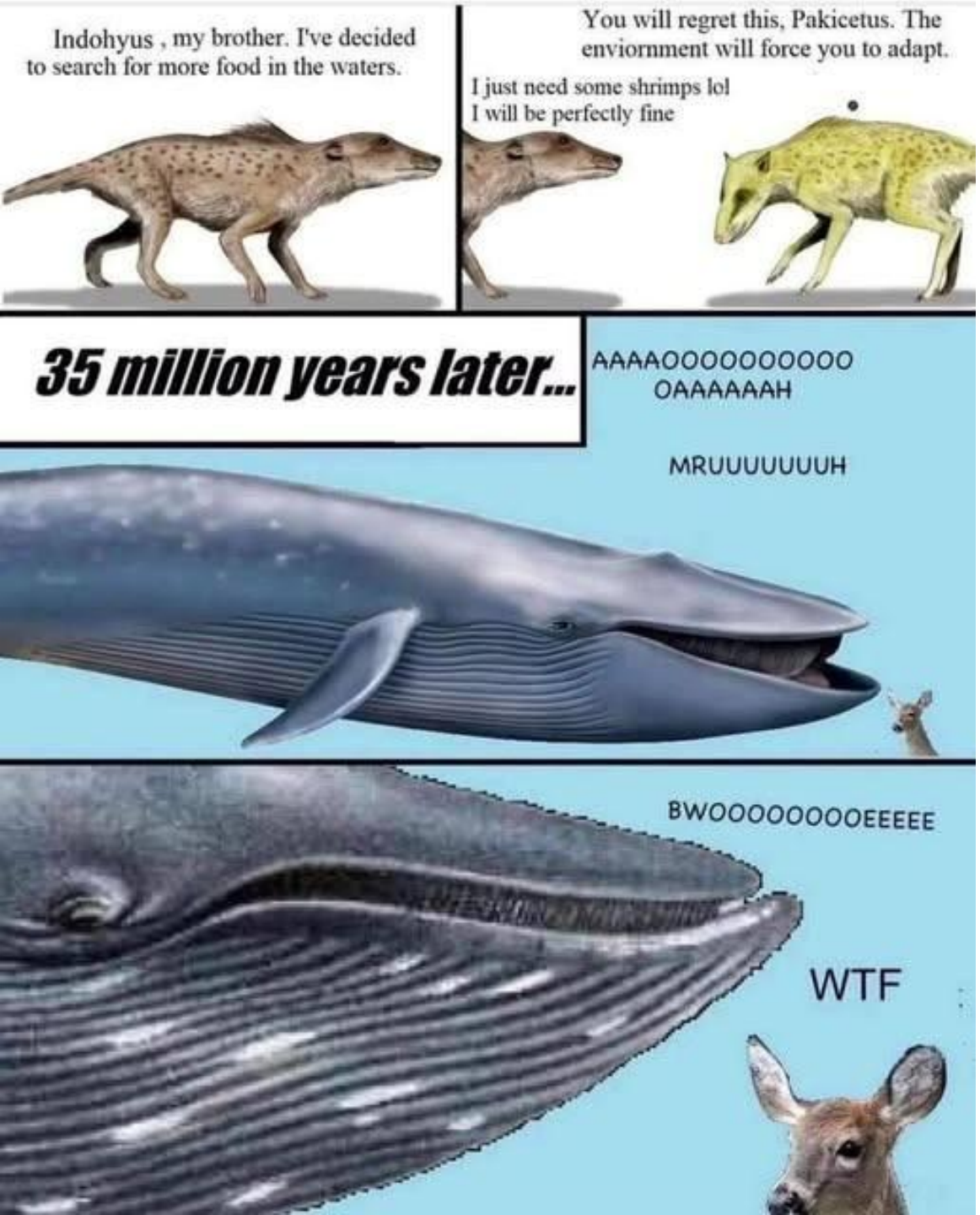


Rose 2006: fig. 14-4

What do you notice about the astragalus in mesonychids, cetaceans, and artiodactyls?



Rose 2006: fig. 14-8 (reversed)



Interesting Fact:

The only 4 mammals to have menopause are:

- Homo sapiens
- Short-fin pilot whales
- Orcinus Orca
- Belugas & Narwhals

S. Ellis et al. 2024, https://doi.org/10.1038/s41586-024-07159-9
L. J. N. Brent et al. 2015, https://doi.org/10.1016/j.cub.2015.01.037
H. Whitehead 2015, https://doi.org/10.1016/j.cub.2015.02.002

There are several papers out there discussing the similarities between human and orca social structures and dynamics and how that influenced the evolution of menopause.

In orcas specifically, it's likely that the selective benefit for a female providing indirect fitness benefits (such as ecological knowledge/teaching) increases the longer they live.

This is because the relatedness within the group increases with age and thus it becomes a greater benefit to a given female if her children and grandchildren are successful rather than having offspring of her own. This is likely why orcas are matriarchal.

However, elephants are matriarchal but do not have menopause, this is likely because the males leave the herd, while in orca they don't. This reduces the indirect fitness benefit upon the matriarch making it no longer worth substituting reproduction for indirect fitness benefits.